# LOB 1003 - Cálculo 1

Lista de exercícios 2 - Parte 2

**1.**

a) $\dfrac{dy}{dx} = \dfrac{-y^2 - 2xy}{x^2 + 2yx}$

b) $\dfrac{dy}{dx} = \dfrac{1 - 2y}{2x + 2y - 1}$

c) $\dfrac{dy}{dx} = \dfrac{-2x^3 + 3x^2y - xy^2 + x}{y - x^3 + x^2y}$

d) $\dfrac{dy}{dx} = \dfrac{1}{y}\dfrac{1}{(x+1)^2}$

e) $\dfrac{dy}{dx} = -\dfrac{(1 - \sqrt{x})}{\sqrt{x}}$

f) $\dfrac{dy}{dx} = \cos^2 y$

g) $\dfrac{dy}{dx} = \dfrac{2e^{2x}}{3\cos(x + 3y)} - \dfrac{1}{3}$

h) $\dfrac{dy}{dx} = \dfrac{-y^2}{y \operatorname{sen}(1/y) + yx - \cos(1/y)}$

i) $\dfrac{dy}{dx} = \dfrac{-y}{x}$

**2.**

a) $y' = \dfrac{-x}{y}, \quad y'' = -\dfrac{1}{y^3}$

b) $y' = \dfrac{xe^{x^2} + 1}{y}, \quad y'' = \dfrac{(y^2 + 2x^2y^2 - 2x)e^{x^2} - 1 - x^2e^{2x^2}}{y^3}$

c) $y' = \dfrac{\sqrt{y}}{1 + \sqrt{y}}, \quad y'' = \dfrac{1}{2(1 + \sqrt{y})^3}$

d) $y' = \dfrac{-x^2}{y^2}, \quad y'' = \dfrac{-2xy^3 - 2x^4}{y^5}$

**3.**

a) $y' = \dfrac{\sqrt{x(x+1)}}{2}\left(\dfrac{1}{x}+\dfrac{1}{x+1}\right)$

b) $y' = \dfrac{1}{2}\sqrt{\dfrac{t}{t+1}}\left(\dfrac{1}{t}-\dfrac{1}{t+1}\right)$

c) $y' = \sqrt{\theta+3}\,\text{sen}\,\theta\left(\dfrac{1}{2(\theta+3)}+\text{cotg}\,\theta\right)$

d) $y' = 3t^2+6t+3$

e) $y' = \left(\dfrac{\theta+5}{\theta\cos\theta}\right)\left(\dfrac{1}{\theta+5}-\dfrac{1}{\theta}+\text{tg}\,\theta\right)$

f) $y' = \dfrac{x\sqrt{x^2+1}}{(x+1)^{2/3}}\left(\dfrac{1}{x}+\dfrac{x}{x^2+1}-\dfrac{2}{3(x+1)}\right)$

g) $y' = \dfrac{1}{3}\sqrt[3]{\dfrac{x(x-2)}{x^2+1}}\left(\dfrac{1}{x}+\dfrac{1}{x-2}-\dfrac{2x}{x^2+1}\right)$

**4.**

a) $\dfrac{\sqrt{2}}{2}$

b) $-\dfrac{\sqrt{3}}{3}$

c) $\dfrac{4+\sqrt{3}}{2\sqrt{3}}$

d) $1$

e) $-\sqrt{2}$

**5.**

a) $y' = \dfrac{-2x}{\sqrt{1+x^4}}$

b) $y' = \sqrt{\dfrac{2}{1-2t^2}}$

c) $y' = \dfrac{s}{|2s+1|\sqrt{s^2+s}}$

d) $y' = \dfrac{-2}{|x^2+1|\sqrt{x^2+2}}$

e) $y' = \dfrac{-1}{\sqrt{1-t^2}}$

f) $y' = \dfrac{-1}{2\sqrt{t}(1+t)}$

g) $y'\dfrac{1}{\text{tg}^{-1}x}\dfrac{1}{1+x^2}$

h) $y' = \dfrac{-1}{\sqrt{e^{2t}-1}}$

i) $y' = \dfrac{-2s^2}{\sqrt{1-s^2}}$

j) $y' = 0$

k) $y' = \text{ sen }^{-1} x$

**8.**

a) $f'(x) = x \cosh x$

b) $h'(x) = \text{tgh } x$

c) $y' = 3e^{\cosh 3x} \text{senh } 3x$

d) $f'(t) = -2e^t \text{sech}^2 e^t \text{tgh } e^t$

e) $G'(x) = \dfrac{2 \text{ cosech } x \text{ cotgh } x}{(1 + \text{cosech } x)^2}$

f) $y' = \dfrac{1}{\sqrt{x-1}} \dfrac{1}{2\sqrt{x}}$

g) $y' = \text{senh}^{-1} \left( \dfrac{x}{3} \right)$

h) $y' = - \text{ cosec} x$

**9.** $\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$.

**10.**

a) $\dfrac{dV}{dt} = \pi r^2 \dfrac{dh}{dt}$

b) $\dfrac{dV}{dt} = 2\pi h r \dfrac{dr}{t}$

c) $\dfrac{dV}{dt} = \pi r^2 \dfrac{dh}{dt} + 2\pi h r \dfrac{dr}{dt}$

**11.**

a) 1 volts/s

b) -1/3 A/s

c) $\dfrac{dR}{dt} = \dfrac{1}{I} \left( \dfrac{dV}{dt} - \dfrac{V}{I} \dfrac{dI}{dt} \right)$

d) 3/2 ohms/s

**12.**

a) $\dfrac{ds}{dt} = \dfrac{x}{\sqrt{x^2 + y^2}} \dfrac{dx}{dt}$

b) $\dfrac{ds}{dt} = \dfrac{1}{\sqrt{x^2 + y^2}} \left( x \dfrac{dx}{dt} + y \dfrac{dy}{dt} \right)$

c) $\dfrac{dx}{dt} = \dfrac{-y}{x}\dfrac{dy}{dt}$

**13.**

a) $L(x) = 10x - 13$

b) $L(x) = 2$

c) $L(x) = x - \pi$

d) $L(x) = 1 + kx$

**14.**

a) $dy = \left(3x^2 - \dfrac{3}{2\sqrt{x}}\right) dx$

b) $dy = \left(\dfrac{2-2x^2}{(1+x^2)^2}\right) dx$

c) $dy = \left(\dfrac{1-y}{3\sqrt{y}+x}\right) dx$

d) $dy = \dfrac{5}{2\sqrt{x}} dx$

e) $dy = 4x^2 \sec^2\left(\dfrac{x^3}{3}\right) dx$

f) $dy = 2(x \operatorname{sen} 2x + x^2 \cos 2x) dx$

g) $dy = \dfrac{1}{2\sqrt{t}} \sec^2 \sqrt{t}\ \ dt$

h) $dy = \dfrac{3 \operatorname{cosec}\,(1-2\sqrt{x})\ \operatorname{cotg}\,(1-2\sqrt{x})}{\sqrt{x}} dx$

i) $dy = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}} dx$

j) $dy = \dfrac{2x}{1+x^2} dx$

k) $dy = \dfrac{2xe^{x^2}}{1+e^{2x^2}} dx$

l) $dy = \dfrac{-1}{\sqrt{e^{-2x}-1}} dx$

m) $dy = \dfrac{e^{x/10}}{10} dx$

n) $dy = \dfrac{x}{\sqrt{3+x^2}} dx$